

Student	Ramasai Behara
Supervisor	Maria Mwanje
Date and time of meeting	17 July 2026, 13:00 PM
State of work	Finalising Chapter 3 (Methodology) and progressing Chapter 4 (Implementation). Novel contribution has been reframed, per supervisor feedback, to investigate whether SMOTE-generated synthetic failure samples distort SHAP feature importance rankings, measured via Spearman rank correlation against bearing (CWRU) and turbofan (NASA C-MAPSS) failure physics. Four imbalance-handling conditions (no handling / SMOTE / ADASYN / class-weighted loss) are being benchmarked across four model architectures (Random Forest, SVM, LSTM, Attention-LSTM) with Bayesian hyperparameter optimisation. Chapter 4 currently covers library/version documentation, global random seed configuration, C-MAPSS preprocessing with documented class imbalance ratios, PyTorch model classes, and the KernelSHAP/TreeSHAP implementations. Work is being tracked against a 13-week Gantt chart (1 June - 28 August 2026) with five milestones.
Is the work on track?	Yes - on schedule against the Gantt chart milestones, with submission due 28 August 2026.
Does the student take initiative to contact the supervisor regularly?	Yes
Is the student aware of opportunities to give – and receive – feedback?	Yes
Have arrangements for supervisor absence during the summer been communicated?	Yes, agreed to have a MS Teams meeting every two weeks
Supervisor's concerns (if any)	Currently no concerns. The student is to focus on the implementation for the next couple of weeks.



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COMP40311 Major Project --- Supervision Record Form

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Supervisor's suggestions (related to hardware/software implementation, testing, collection of primary data, write-up, etc.)	Advised the student to double check the current state of art applications or methodologies within the research area to ensure that the claims regarding the new ideas are valid.
Supervisor's signature	<i>Maria Mwanje</i>
Student's signature	